

Double-Angle Formulas and Trigonometric Equations

ANSWER KEY



Double-angle formulas

- 1 Given that $\cos \theta = -\frac{3}{5}$ and θ lies in Quadrant II, find:
- a** $\sin 2\theta = -\frac{24}{25}$ **b** $\cos 2\theta = -\frac{7}{25}$ **c** $\tan 2\theta = \frac{24}{7}$
- 2 Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.
- $\cos^2 x = \frac{1 + \cos 2x}{2}$ and $\sin^2 x = \frac{1 - \cos 2x}{2}$.

Solving trigonometric equations

- 3 Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.
- $\sin x = \frac{1}{2}$, so $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$.
- 4 Solve $2\cos^2 x - \cos x - 1 = 0$ on the interval $[0, 2\pi)$.
- Factor: $(2\cos x + 1)(\cos x - 1) = 0$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$; from $\cos x = 1$: $x = 0$. Solution set: $\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\}$.
- 5 Solve $\sin 2x = \sin x$ on the interval $[0, 2\pi)$.
- $2\sin x \cos x - \sin x = 0$, so $\sin x(2\cos x - 1) = 0$. From $\sin x = 0$: $x = 0, \pi$; from $\cos x = \frac{1}{2}$: $x = \frac{\pi}{3}, \frac{5\pi}{3}$. Solution set: $\{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}\}$.
- 6 Solve $\tan^2 x = 3$ on the interval $[0, 2\pi)$.
- $\tan x = \pm \sqrt{3}$, so $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

Half-angle formulas

- 7 Use a half-angle formula to find the exact value of $\cos 15^\circ$ (using $15^\circ = \frac{30^\circ}{2}$).
- $\cos 15^\circ = \sqrt{\frac{1 + \cos 30^\circ}{2}} = \sqrt{\frac{1 + \frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2 + \sqrt{3}}{4}} = \frac{\sqrt{2 + \sqrt{3}}}{2}$ (positive since 15° is in Quadrant I).
- 8 Given $\cos \theta = \frac{7}{25}$ with $0 < \theta < \frac{\pi}{2}$, find $\sin \frac{\theta}{2}$ exactly.
- $\sin \frac{\theta}{2} = \sqrt{\frac{1 - \cos \theta}{2}} = \sqrt{\frac{1 - 7/25}{2}} = \sqrt{\frac{18/25}{2}} = \sqrt{\frac{9}{25}} = \frac{3}{5}$ (positive since $\frac{\theta}{2}$ is in Quadrant I).

Equations requiring double-angle substitution

- 9 Solve $\cos 2x + \sin x = 0$ on the interval $[0, 2\pi)$, using $\cos 2x = 1 - 2\sin^2 x$.
- $1 - 2\sin^2 x + \sin x = 0$, so $2\sin^2 x - \sin x - 1 = 0$: $(2\sin x + 1)(\sin x - 1) = 0$. From $\sin x = -\frac{1}{2}$: $x = \frac{7\pi}{6}, \frac{11\pi}{6}$; from $\sin x = 1$: $x = \frac{\pi}{2}$. Solution set: $\{\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}\}$.

- 10 Solve $\sin 2x - \cos x = 0$ on the interval $[0, 2\pi)$.

$2\sin x \cos x - \cos x = 0$, so $\cos x(2\sin x - 1) = 0$. From $\cos x = 0$: $x = \frac{\pi}{2}, \frac{3\pi}{2}$; from $\sin x = \frac{1}{2}$: $x = \frac{\pi}{6}, \frac{5\pi}{6}$.
Solution set: $\{\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}\}$.

Applications of double-angle formulas

- 11 A projectile's range is given by $R(\theta) = \frac{v^2 \sin 2\theta}{g}$. Using the double-angle formula, explain why the maximum range occurs at $\theta = 45^\circ$.

$R(\theta) = \frac{v^2}{g} \sin 2\theta$ is maximized when $\sin 2\theta = 1$, i.e. $2\theta = 90^\circ$, so $\theta = 45^\circ$.

- 12 Given $\sin \theta = \frac{1}{3}$ with θ in Quadrant I, find $\cos 2\theta$ and $\sin 2\theta$ exactly.

$\cos \theta = \sqrt{1 - 1/9} = \frac{2\sqrt{2}}{3}$. $\cos 2\theta = 1 - 2\sin^2 \theta = 1 - \frac{2}{9} = \frac{7}{9}$. $\sin 2\theta = 2\sin \theta \cos \theta = 2 \cdot \frac{1}{3} \cdot \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{9}$.

More equation practice

- 13 Solve $4\sin^2 x - 3 = 0$ on the interval $[0, 2\pi)$.

$\sin^2 x = \frac{3}{4}$, so $\sin x = \pm \frac{\sqrt{3}}{2}$: reference angle $\frac{\pi}{3}$, giving $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.

- 14 Solve $\cos 2x = \cos x$ on the interval $[0, 2\pi)$, using $\cos 2x = 2\cos^2 x - 1$.

$2\cos^2 x - 1 = \cos x$: $2\cos^2 x - \cos x - 1 = 0$, so $(2\cos x + 1)(\cos x - 1) = 0$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$; from $\cos x = 1$: $x = 0$. Solution set: $\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\}$.